

MPG No. 11.1.35 4/m [Type I, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = M_{pz}^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = -M_{faz}^{(1)} + M_{pz}^{(1)}$$

$$M_{zzz} = 2M_{faz}^{(1)} + 3M_{pz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ M_{yzz} & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & -M_{yzz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2M_{pz}^{(2)}$$

$$M_{yzz} = 3T_{du}^{(1)}$$

$$\frac{M_{xxz}}{2} = M_{pz}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzz} & M_{yzy} & 0 \\ -M_{yzy} & M_{yzz} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzz} = -T_{du}^{(2)} + T_s^{(1)}$$

$$M_{yzy} = M_{pz}^{(3)}$$

$$M_{xyz} = 2T_{du}^{(2)} + T_s^{(1)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).